COMP 3500SEF · Final Report

Warehouse Logistics Management System
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1. Scope

The goals are traceable order decisions and a role-separated workflow. Customers order and track; staff confirm, pick, ship and assign; drivers simulate assigned deliveries; administrators report. The local application uses Next.js, TypeScript, Prisma and SQLite. GPS, scanning, telephony, payment and carrier integration are outside scope. Deployment is local-only; data integrity is enforced by transactional guards (§5).

2. Development process

Increments have triggers and verified outcomes. Increments cover product and specification changes and recorded process milestones.

Table A. Increment, trigger and outcome

Increment and date	Trigger	Outcome
Planning, 9 and 11 Oct 2026	Project selection	Responsibilities and contracts recorded (register)
17 Oct test-led review	Assess order and stock functions	Defects recorded in the register; led to the confirmation transaction (§5)
Confirmation, 26–28 Oct 2026	Failed endpoint and missing stock ledger (register)	Transaction verified (orders tests)
Integrated application, 1 Nov 2026	Integrate foundation	Customer, staff and admin workflows
Driver and privacy, 2 Nov 2026	Delivery and data minimisation	Assigned tasks and data-minimisation checks (tests)
Documentation hardening, 10–11 Nov 2026	Model and evidence review	Diagrams and traceability aligned (traceability)
Export simplification, 11 Nov 2026	11 Nov scripted walkthrough (issue #15)	Selected-date export in three actions; regression passed
Availability tests, 11 Nov 2026	Regression gap in integration-health checks	Suite increased to 38 (unit.log)

3. Requirements

Functional requirements and verification

FR	The system shall	Verification
FR-01	create orders with product lines and totals	Test — creation and stock
FR-02	enforce lifecycle and stock reservation or release	Test — lifecycle and stock
FR-03	number orders and shipments and show owner tracking	Test — identifiers and tracking
FR-04	show inventory, replenishment, ledger (inventory movement log) and low-stock alerts	Test — inventory and alerts
FR-05	record event notifications and allow reading	Test — notifications and reads
FR-06	provide admin aggregates and CSV export	Test — aggregates and export
FR-07	authenticate and enforce role and object access	Test — authentication and boundaries
FR-08	assign shipped orders and simulate delivery	Test — assignment and delivery

NFR-01 performance (API p95 ≤2 s, first contentful paint ≤3 s); NFR-02 integration health (green build–test–browser runs recorded per increment); NFR-03 least privilege and data minimisation — role gates on every mutating endpoint; staff order responses omit customer identity and delivery address (privacy tests); NFR-04 ≤3 actions per scripted operation; NFR-05 process quality — CI-verified increments (workflow).

Table B. Risk coverage (detailed mapping)

Risk	Addressed by	Evidence
RK-01 scheduling slip	Increment register	scheduling plan
RK-02 contract rework	FR-02, NFR-05	transition tests
RK-03 test-week capacity	Priority register	scope priorities
RK-04 setup delay	NFR-05	CI workflow
RK-05 demo data	FR-01/04/06, NFR-05	seed regression test
RK-06 assignments and test squeeze	Risk plan	assessment schedule
RK-07 stack unfamiliarity	NFR-05 support	setup guide
RK-08 report and video compete for time	Risk plan	report and video plan

4. System models

Local transactions preserve consistency without splitting services; state and stock guards enforce rules. The four layers — Role-specific pages, HTTP routes and authentication, Order/inventory/driver business handlers, Prisma/SQLite persistence — each isolate one concern.

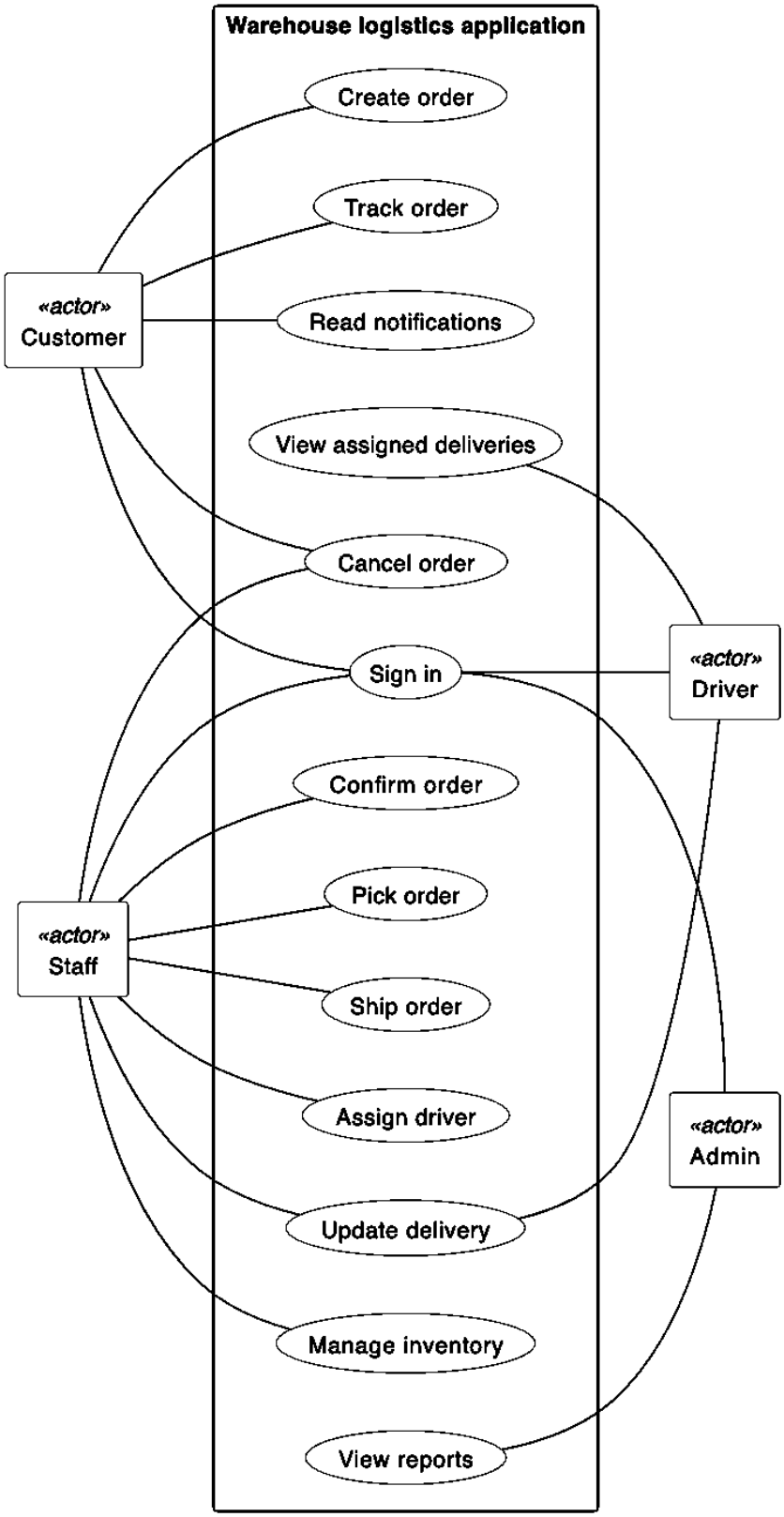


Figure 1. Which roles perform each function? Source.

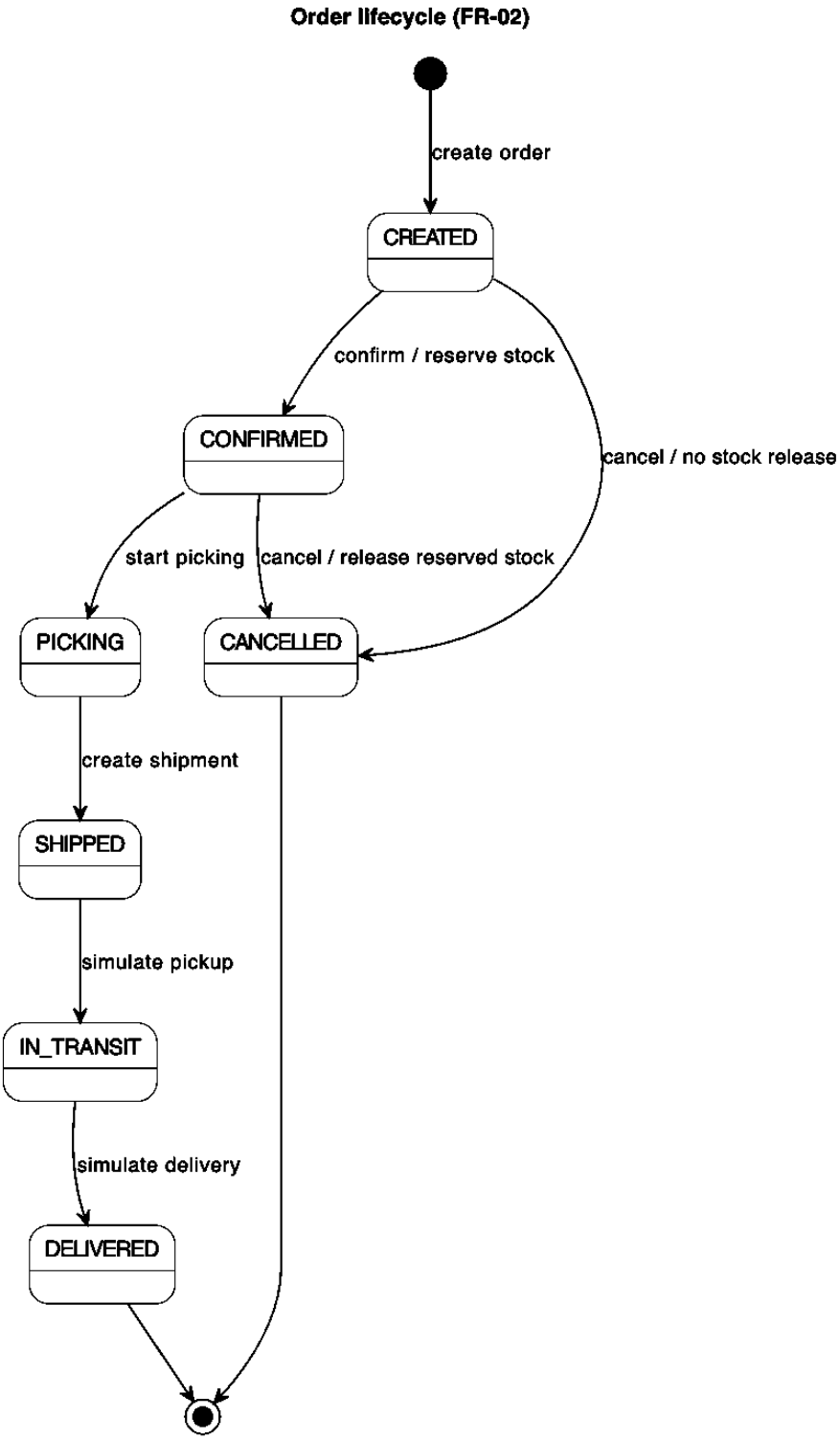


Figure 2. Which order transitions and stock effects are valid? Source.

Each item references one product; each order requires at least one item; each order has 0..1 shipment; each product has 0..* inventory logs.

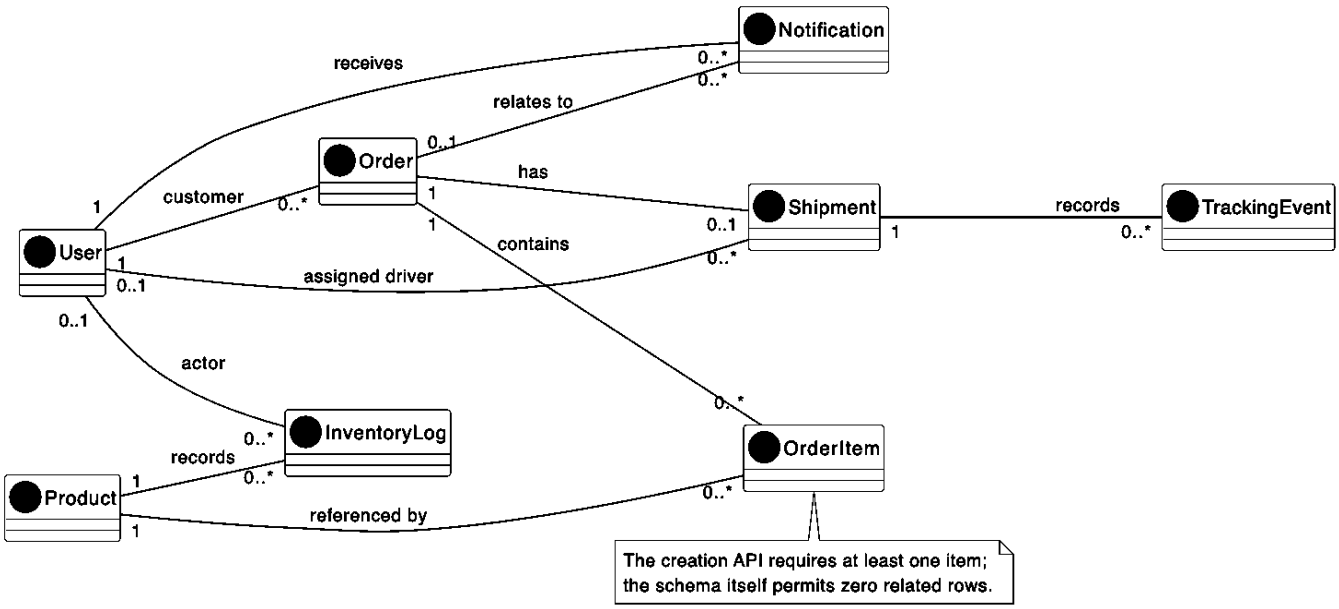


Figure 3. Data relationships: each order requires at least one item, enforced at the creation API. Source.

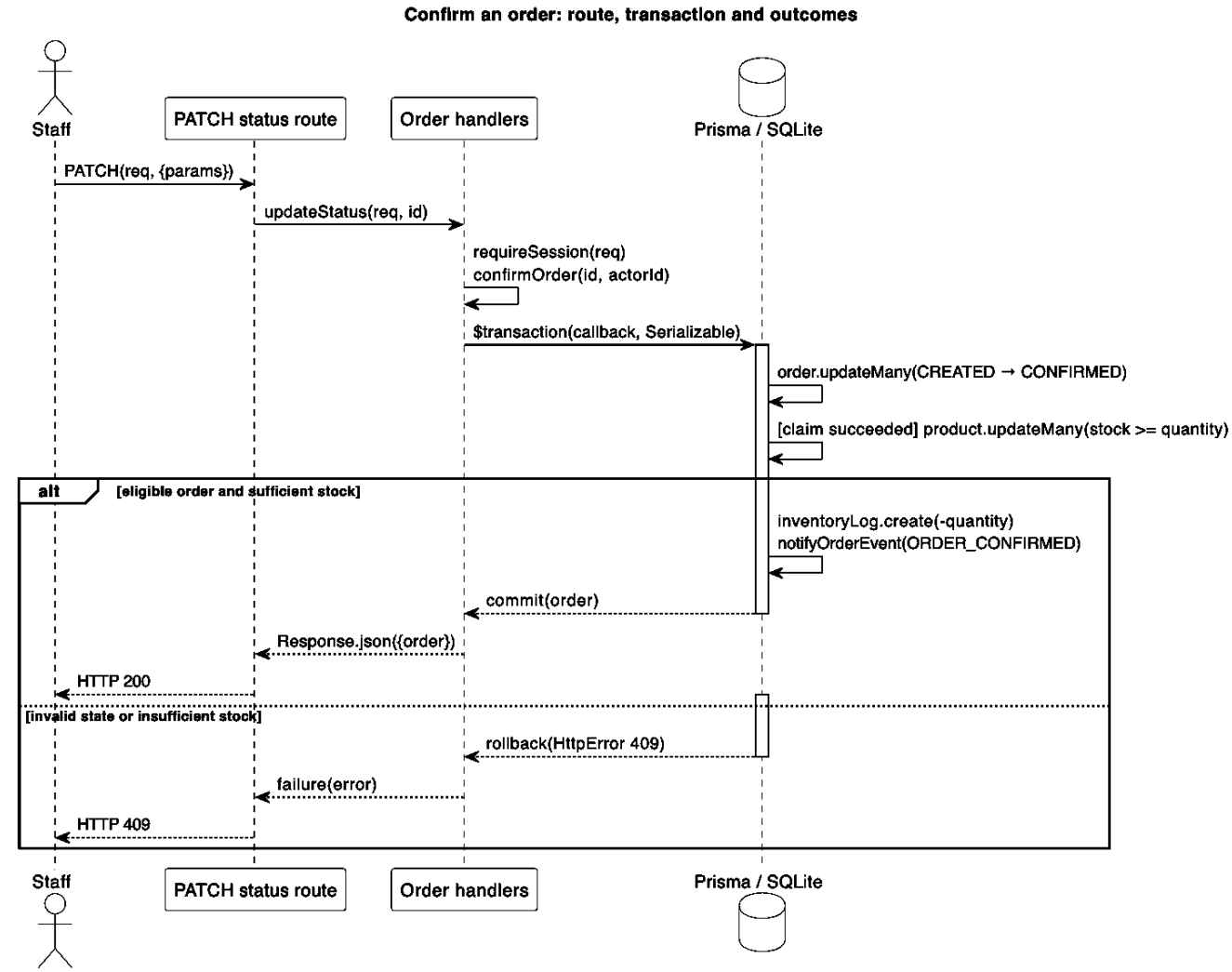


Figure 4. How does confirmation commit or roll back? Source.

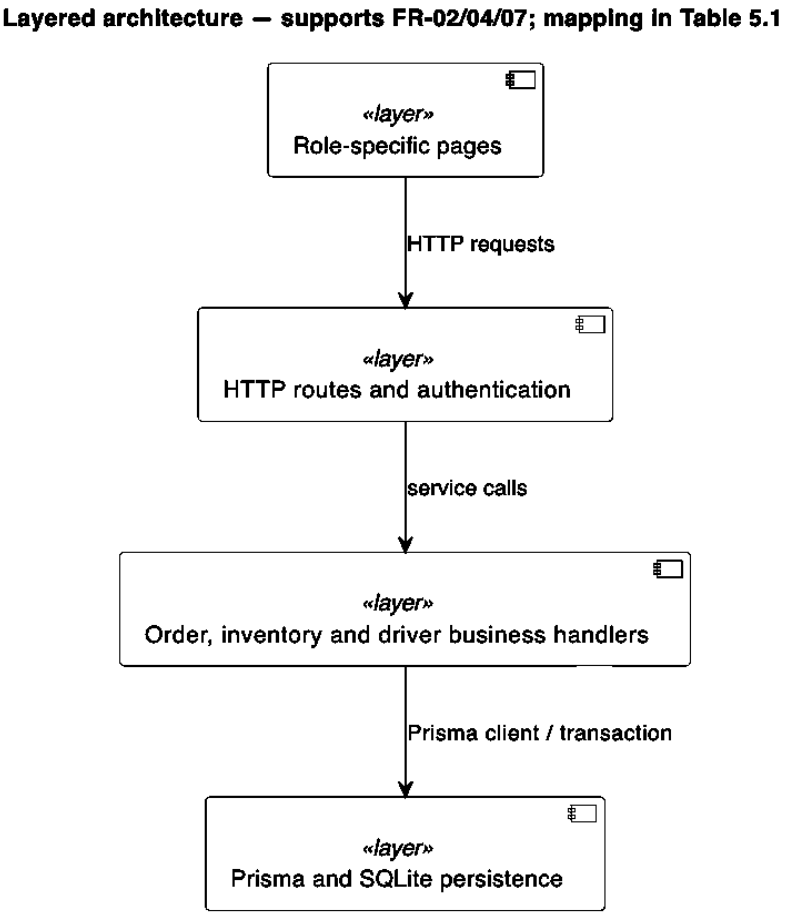


Figure 5. How do four layers connect? Source.

5. Implementation

Stock is checked at creation and reserved at confirmation; the guards below and the confirmation transaction (order-api.ts:245–293) enforce both steps.

Cancellation:

Role	State	Effect
Customer	CREATED	cancel, no restock
Customer	CONFIRMED	cancel, restock
Staff	CONFIRMED	cancel, restock
Both	PICKING onwards	rejected

Design	Implementation	Verification
Guards: CREATED and sufficient stock	order-api.ts:250–253, 276–280	Orders
Confirmation transaction: serializable state, stock, ledger and notification writes	order-api.ts:245–293	Orders
driverId: assign only SHIPPED; retained after delivery	driver-api.ts:44–51; transition	Driver
FR-03 numbering and tracking	create-order.ts:91, shipment-api.ts:62	Logistics chain
FR-04 inventory ledger	order-api.ts:281, 398	Inventory
FR-05 notifications	notify.ts:19	Notifications
FR-06 report export	admin/page.tsx:24, 44	Reports

6. Verification and validation

Verification checks requirements against the implementation; validation exercises the delivered workflows through the interface (§8). The regression log and CI show 38 unit, 140 integration, 2 delivery checks (assignment and delivery simulation) (driver tests) and Playwright 26 browser checks.

Insufficient stock returns HTTP 409 with unchanged state. In a local-loopback measurement recorded on 11 November 2026, API p95 was 3.2–13.6 ms and first-contentful-paint p95 was 64–96 ms, against NFR-01 targets of at most 2 s and 3 s respectively. NFR-01 is met with more than an order of magnitude of headroom on both metrics. NFR-02 is evidenced by the recorded green build–test–browser runs (observations).

7. Evolution

The 17 Oct test-led review found a failed confirmation endpoint and a missing stock ledger; the confirmation transaction and its retest followed (26–28 Oct). The 10–11 Nov documentation review corrected the diagrams and aligned traceability with the corrected models. Two integration-health regression tests added on 11 Nov brought the suite to 38.

8. Validation

The automated walkthrough exercises eleven core operations executed by their owning roles (11 role-executions in total), at most three actions each; every operation has expected outcomes asserted in the walkthrough script, and the Playwright 26 browser checks cover the eleven operations at one to three checks each. Results are logged with the walkthrough record.

9. Feedback and reconciliation

Table C. Feedback to evidence

Feedback	Disposition	Source	Eviden
Diagram bases diverged from the implemented schema	Models corrected	10–11 Nov documentation review	Compli
Stock effects were recorded under FR-04	Moved to FR-02	17 Oct test-led review	Traceal
Trigger and outcome detail was thin	Expanded	10–11 Nov documentation review	Registe
The admin export took four scripted interactions	Re-tested	11 Nov scripted walkthrough	Issue #
Integration-health checks had no regression test	Suite reached 38	regression runs	Test his
NFR criteria assumed a hosted deployment model	Set for local deployment	regression runs	SRS re

10. Evidence

Traceability · Baseline record.

Baseline `6eb01afb1798c7361205b10597d80c0c26e8fdf6` pins the exact code, evidence and logs reviewed in this report, at Repository.